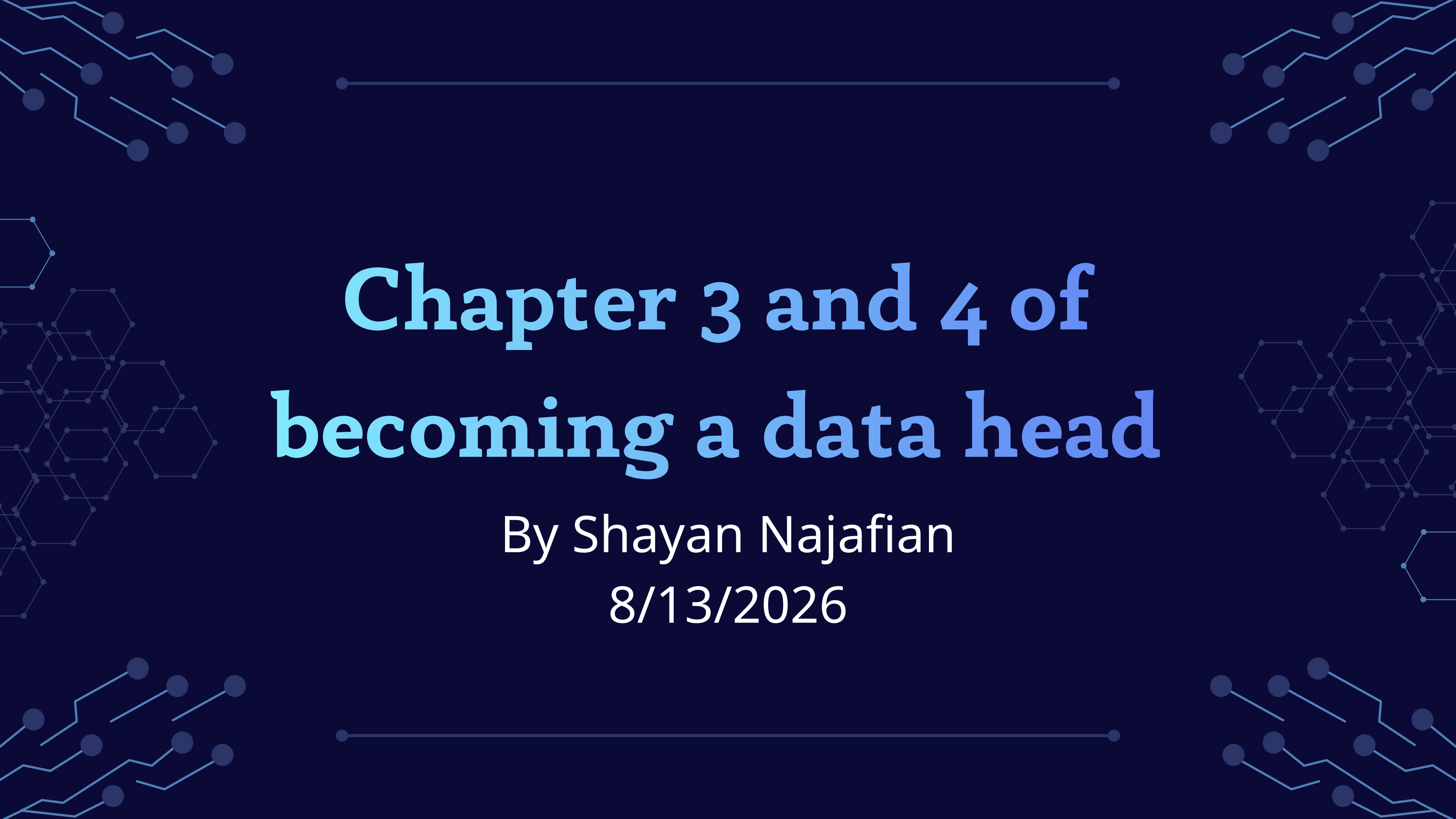
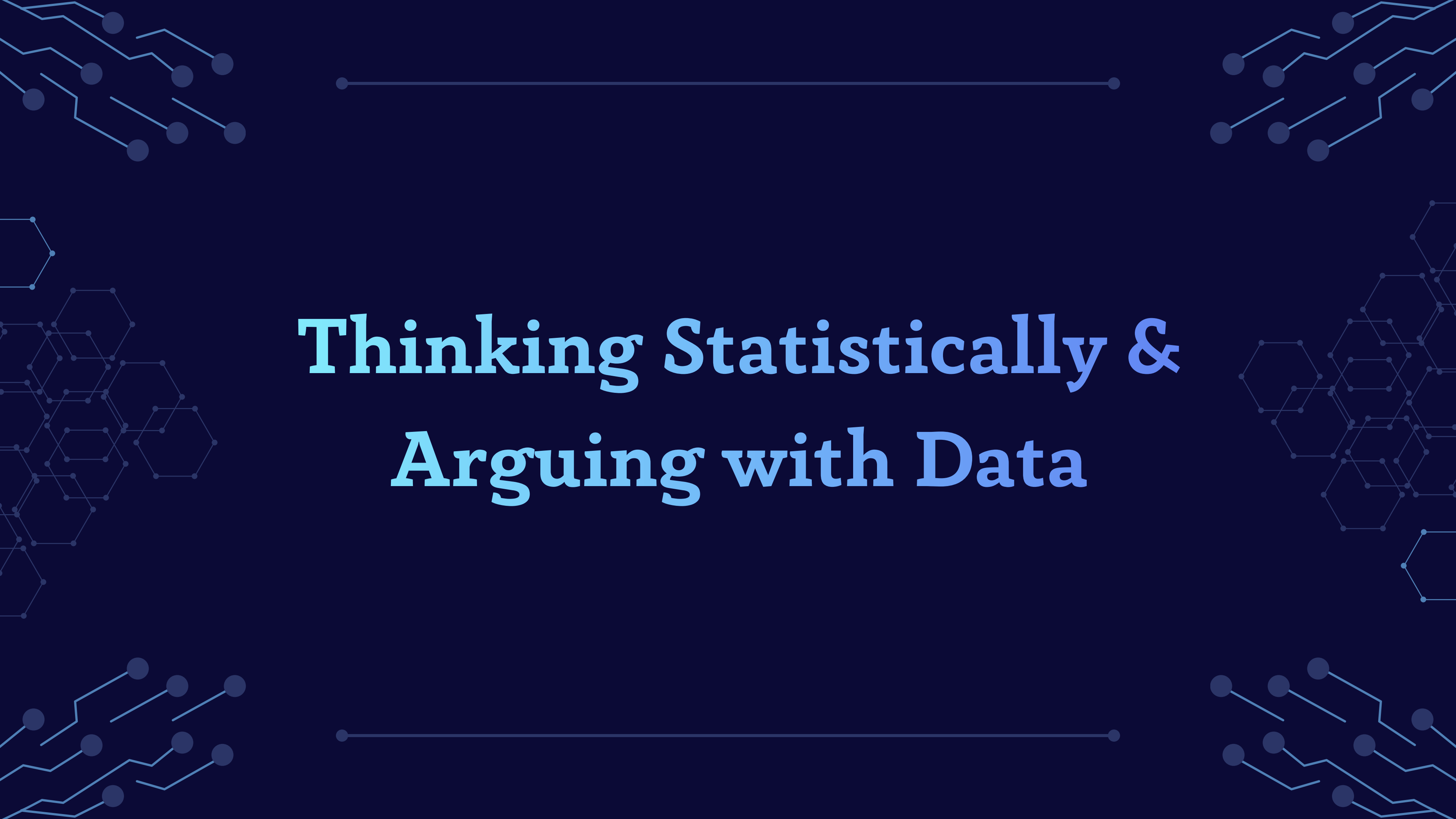




Chapter 3 and 4 of becoming a data head

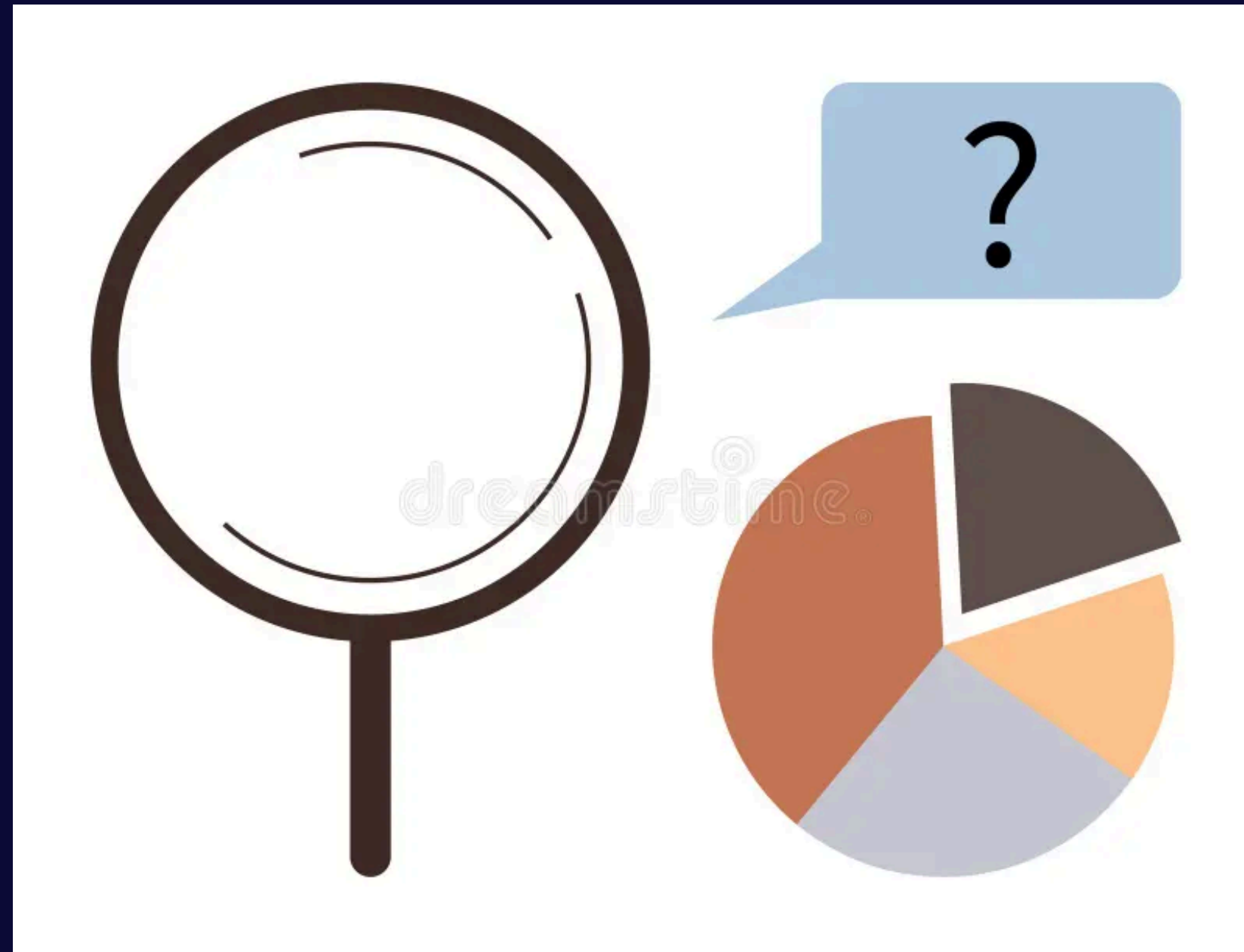
By Shayan Najafian

8/13/2026

The background is a dark blue gradient. In the four corners, there are abstract geometric patterns. The top-left and bottom-left corners feature a network of light blue lines with circular nodes. The top-right and bottom-right corners feature a pattern of light blue hexagons. The center of the image contains the title text in a light blue serif font.

Thinking Statistically & Arguing with Data

The Core Story



The Core Story

1. Is the pattern real, or could it be random variation?
2. Does the data actually represent what we think it represents?

What Is Statistical Thinking?

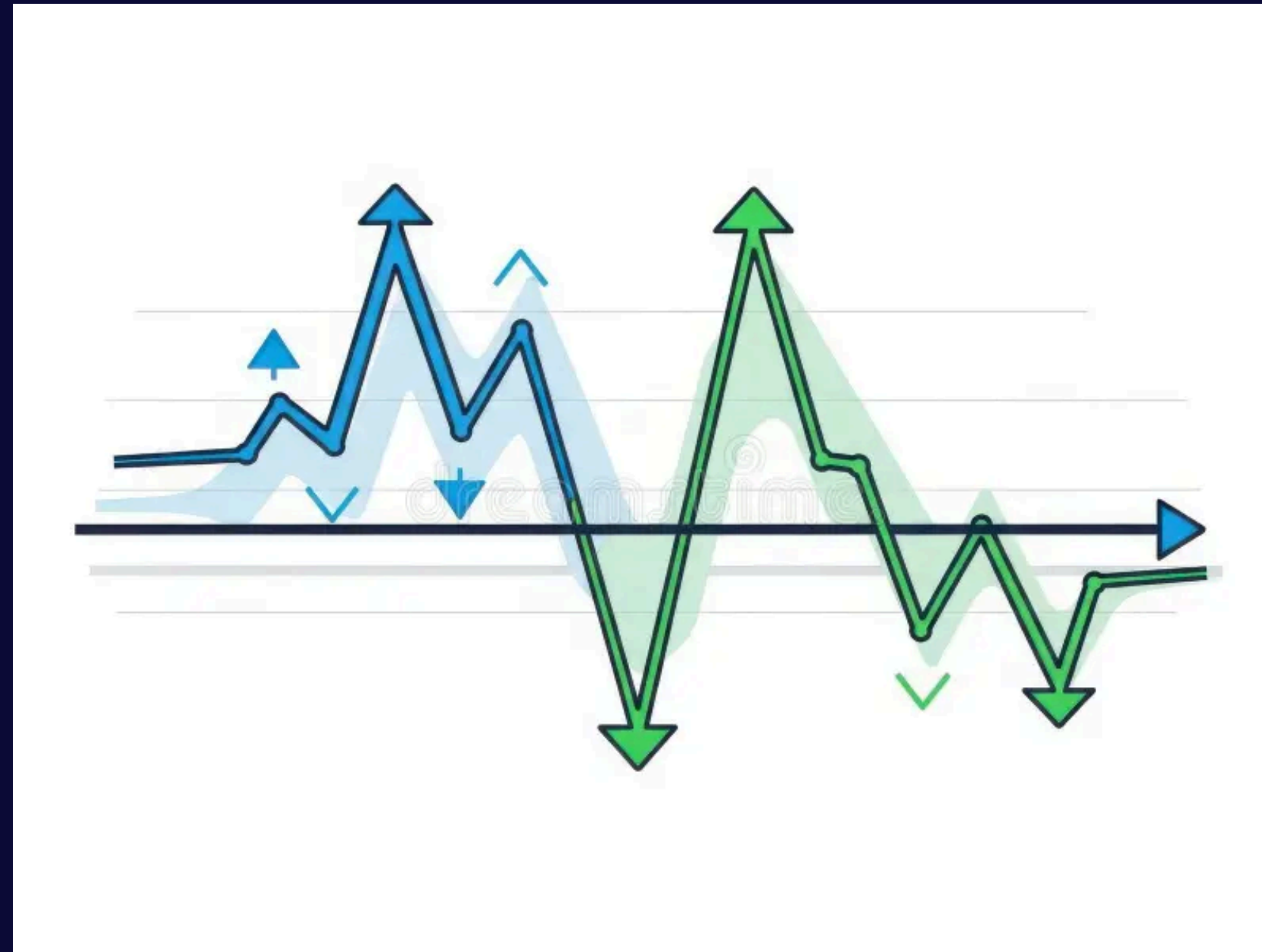
Be curious. Be skeptical. Consider alternative explanations.

A number is evidence—not automatically the truth.

Ask Questions, Don't Just Accept Numbers

Where did this number come from?
How certain is it?
What assumptions are behind it?

There Is Variation in Everything





There Is Variation in Everything

Measurements naturally change.

Sales fluctuate. Surveys fluctuate. Markets fluctuate.
Different does not always mean something changed.



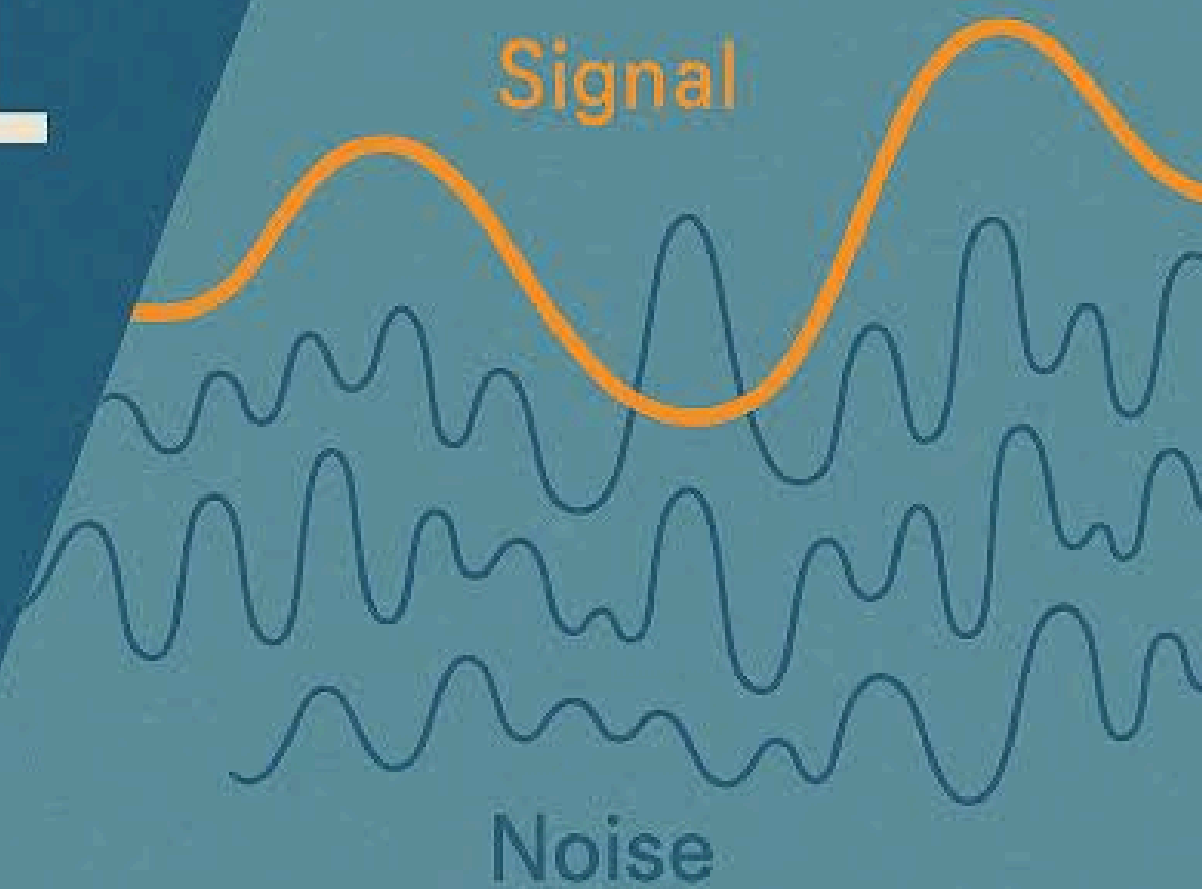
Measurement Variation vs. Random Variation

Measurement variation: caused by how we measure.

Random variation: naturally exists in the process.

Both create uncertainty.

SIGNAL VS NOISE



Signal vs. Noise

Signal = a meaningful underlying change

Noise = variation that happens anyway

Don't Explain Every Peak and Valley

A change happened.

≠

Something caused the change.

First ask:

Could ordinary variation explain it?

Small Samples Create Extreme Results

Small sample → more variation

Large sample → more stability

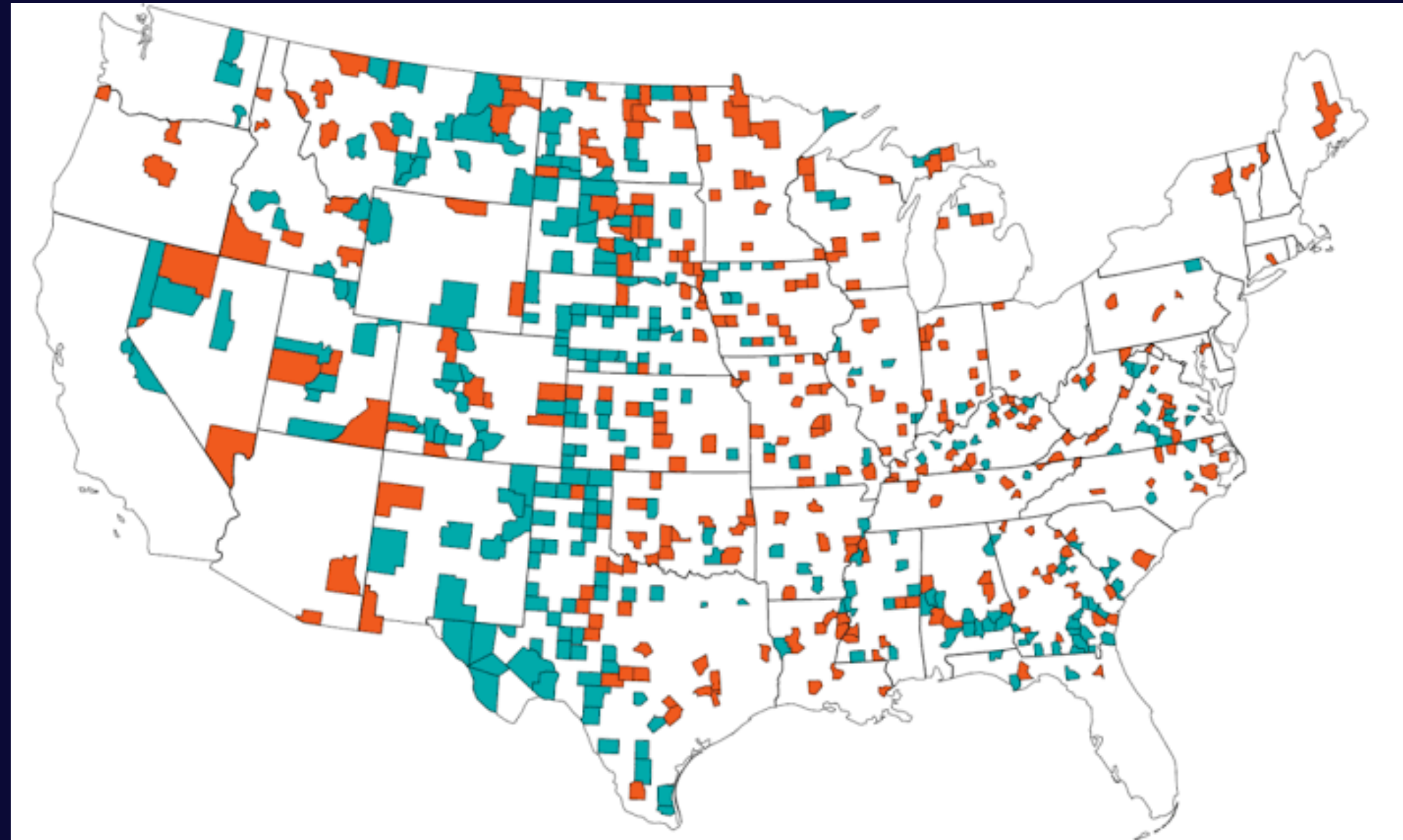
Kidney-Cancer Rates

The highest kidney-cancer rates appeared in small rural counties.

But so did the lowest rates.

Why? Small populations create large variation.

Kidney-Cancer Rates



Kahneman: The Law of Small Numbers

We expect small samples to behave like the population.

But:

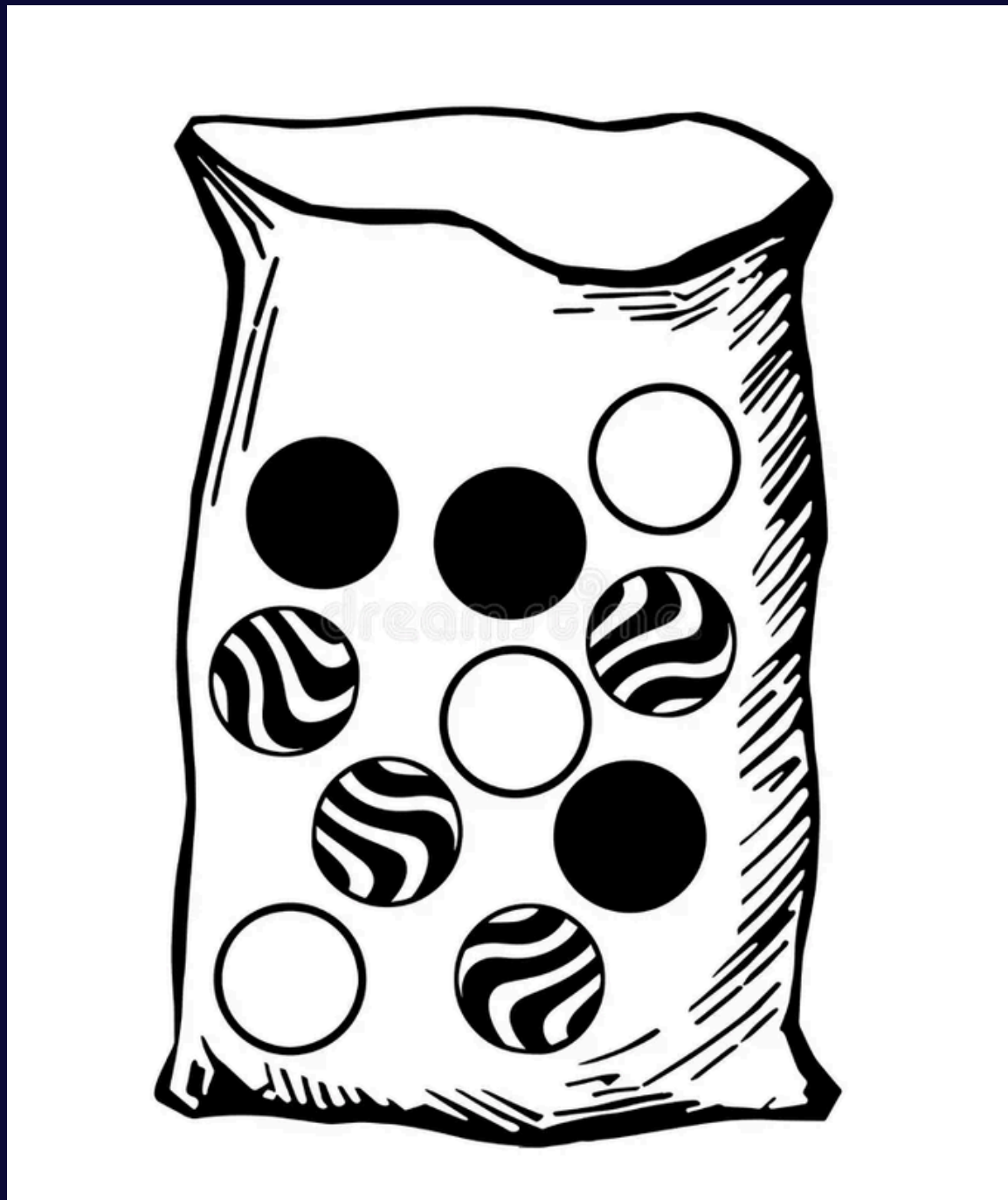
Small samples are much noisier than our intuition expects.

Probability vs. Statistics

Probability: Population $\rightarrow$ Sample

Statistics: Sample $\rightarrow$ Population

Probability drills down. Statistics drills up.



Probability vs. Intuition

80% of dentists agree.

If we randomly choose 5 dentists:

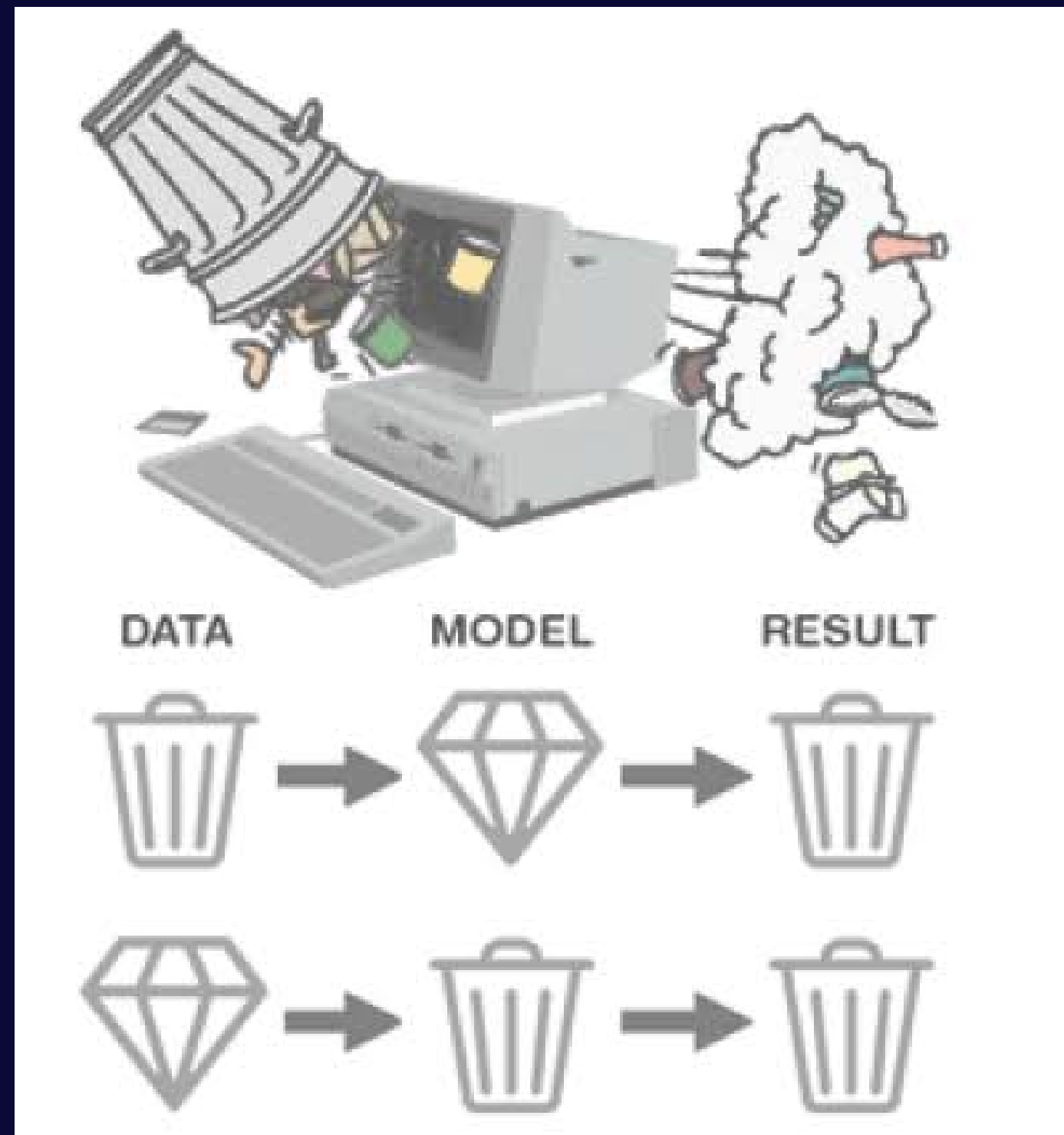
$P(\text{exactly 4 agree}) \approx 41\%$

Not 80%.

Descriptive vs. Inferential Statistics

$$P(X = 4) = \binom{5}{4} (0.8)^4 (0.2) \approx 0.4096$$

Argue with the Data



Argue with the Data

Garbage in → Garbage out

No model, algorithm, or statistical technique can rescue fundamentally bad input data.

Tell Me the Data Origin Story

Before analyzing, ask:
Who collected it?
How was it collected?
Why was it collected?

Representative Data, Bias & Outliers

A large sample is not necessarily a representative sample.
More data reduces random error.
It does not automatically remove bias.

What Data Am I Not Seeing?

Sometimes the most important data is:

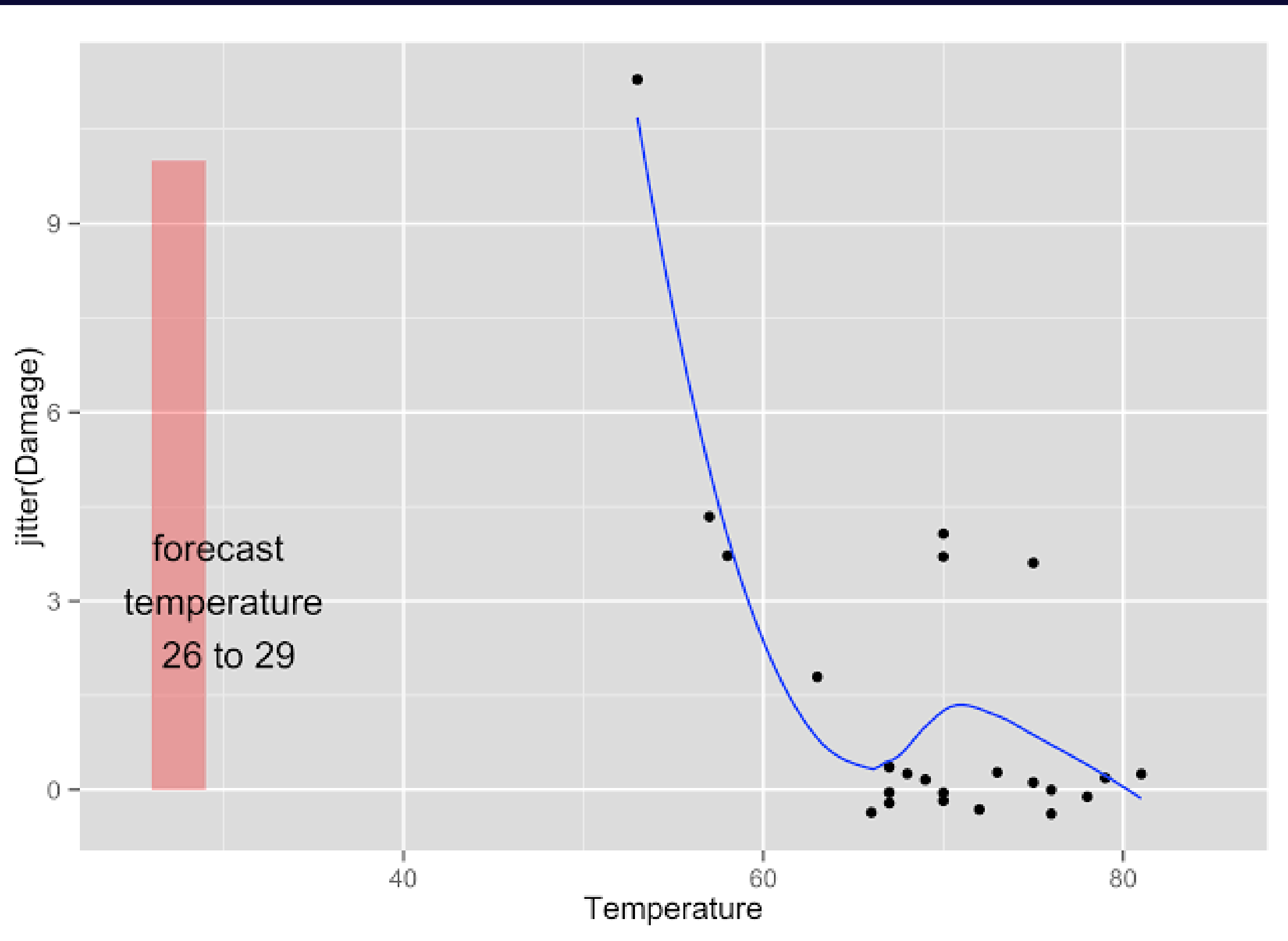
The data that is missing.

Also ask:

Are we measuring reality—or only a proxy for it?

Challenger: What Data Are We Not Seeing?

The analysis showed flights where O-rings failed.
But many flights where they did not fail were missing.
Always ask for the denominator.



Conclusion: Think Like a Data Head

Question the pattern.

Question the sample.

Question the data.

And underneath:

Good analysis starts before the model.



Thank You

Questions & Discussion